

Nasreen Buhn

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EDUCATION

California Polytechnic State University, San Luis Obispo, CA

Bachelor of Science in Biological Sciences, Concentration in Cellular and Molecular Biology, Minors in Computer Science and Bioinformatics, *Cum Laude* June 2025

Relevant Coursework: Deep Learning, Bioinformatics Algorithms, Bioinformatics Capstone I & II

Honors: President's Honors List (2023-2024), Dean's List (6 quarters, 2022-2025)

RESEARCH EXPERIENCE

Gill Lab, Perelman School of Medicine at the University of Pennsylvania

Bioinformatics Research Specialist, December 2025 – Present

Advisors: Saar Gill, Dokyoon Kim, July 2026 – Present

- Developing a computational framework integrating single-cell RNA-seq and ATAC-seq, gene regulatory network inference, knowledge graphs, and deep learning to design minimal synthetic cis-regulatory elements for state-specific expression in terminally exhausted CD8⁺ T cells

Advisor: Saar Gill, December 2025 – July 2026

- Analyzing single-cell RNA-seq, whole-genome sequencing, and longitudinal cytokine data to investigate CAR T-cell treatment response, relapse, and CRS/IEC-HS across patient cohorts.
- Built a bulk RNA-seq survival analysis platform integrating TCGA expression and clinical data to identify prognostic biomarkers in AML and ALL
- Developed an interactive dashboard enabling lab members to independently analyze cytokine data and generate publication-quality figures

California Polytechnic State University, Computer Science & Software Engineering Department

Summer Undergraduate Research Program, College of Engineering, June 2024 – September 2024

Undergraduate Research Assistant, September 2024 – December 2025

Advisor: Jonathan Ventura

- Benchmarked supervised deep learning models for fluorescence microscopy image denoising on a novel dataset, implementing standardized preprocessing and model training procedures to ensure fair cross-model comparison
- Developed a stitching algorithm to reconstruct full-resolution images from overlapping, denoised segments with varying light intensities

Bioinformatics Research Group, California Polytechnic State University, Biological Sciences and Computer Science & Software Engineering Departments

Undergraduate Research Assistant, January 2024 – June 2025

Advisors: Jean Davidson, Paul Anderson

- Collaborated on the design and development of a zero-shot entity resolution methodology assessing the impact of context in NIH-funded chronic lower back pain (CLBP) knowledge graph research
- Evaluated entity pairs to build a ground-truth dataset and helped craft LLM prompting strategy

- Assessed the performance of protein clustering and alignment algorithms for classifying carbonic anhydrase proteins. Applied SVD and k-means clustering, evaluating the impact of sequence length and clustering conditions using silhouette scores. Compared multiple-sequence alignment and multiple-structure alignment methods with phylogenetic tree analysis

PUBLICATIONS

Buhn, N., Adunur, S., Hamilton, J., Levis, S., Hagen, G., & Ventura, J. (2026). Comparison of deep learning approaches for extreme low-SNR image restoration. *GigaScience*, 15, Article giag071.

<https://doi.org/10.1093/gigascience/giag071>

Lin, D., Koenig, C., Kaplan, S., Bittner, M., Paraiso, M., **Buhn, N.**, et al. (2025). Investigating the impact of context on zero-shot entity resolution: Applications in chronic lower back pain. In *2025 8th International Conference on Information and Computer Technologies (ICICT)* (pp. 390–396). IEEE.

<https://doi.org/10.1109/ICICT64582.2025.00067>

Anderson, P., Lin, D., Davidson, J., Migler, T., Ho, I., Koenig, C., Bittner, M., Kaplan, S., Paraiso, M., **Buhn, N.**, et al. (2024). Bridging domains in chronic lower back pain: Large language models and ontology-driven strategies for knowledge graph construction. *Bioinformatics and biomedical engineering: IWBBIO 2024* (Lecture Notes in Computer Science Vol. 14849). Springer. https://doi.org/10.1007/978-3-031-64636-2_2

MANUSCRIPT IN PREPARATION

Standard-of-care chemotherapies combine with CD24 blockade to enhance cancer cell elimination by macrophages. Manuscript in preparation with Pheast Therapeutics.

ORAL PRESENTATIONS

Bittner, M., **Buhn, N.**, et al. (2025, May 15-16). *Bioinformatics capstone 2025: Solving real world genomics problems with biotechnology collaborators: Developing a comparative transcriptomics pipeline to identify key biomarkers for pre-metastatic tumor buds* [Conference Session]. Bailey College Student Research Conference 2025, California Polytechnic State University, San Luis Obispo, CA, USA.

POSTER PRESENTATIONS

Butler, J., **Buhn, N.**, et al. (2025, November 15). *Training the next generation in molecular pathology through industry-academic collaboration: precision tissue enrichment of colorectal tumor buds analyzed by undergraduate bioinformatics students.* Association for Molecular Pathology (AMP) 2025 Annual Meeting and Expo, Boston, MA, USA.

Buhn, N., Adunur, S., Hagen, G., & Ventura, J. (2024, October 18). *Deep learning for microscope image denoising.* Summer Undergraduate Research Program Symposium 2024, California Polytechnic State University, San Luis Obispo, CA, USA. https://digitalcommons.calpoly.edu/ceng_surp/83/

Koenig, C., Bittner, M., Kaplan, S., Paraiso, M., **Buhn, N.**, et al. (2024, May 16-17). *Novel approaches to literature analysis and entity resolution for chronic lower back pain; Knowledge graphs and large language models.* Bailey College Student Research Conference 2024, California Polytechnic State University, San Luis Obispo, CA, USA.

Buhn, N. (2023, October 23-25). *Real world evidence in FDA drug approvals 2021-2023.* The Organization for Professionals in Regulatory Affairs' (TOPRA) Symposium 2023, Lisbon, Portugal.

TEACHING EXPERIENCE

Parker Institute for Cancer Immunotherapy (PICI), University of Pennsylvania

Teaching Assistant, Bioinformatics in Cancer Immunotherapy Summer Course, Summer 2026

RELATED PROFESSIONAL EXPERIENCE

Pheast Therapeutics, California Polytechnic State University

Bioinformatics Consultant (Contract), August 2025 – December 2025

Advisors: Jean Davidson, Jen Cao

- Evaluated longitudinal treatment responses to a novel drug in combination with small-molecule inhibitors (SMIs) in macrophage–cancer cell co-cultures across cell lines, donors, and doses. Applied DTW-based unsupervised clustering to identify distinct treatment-response profiles

QuantumCyte, California Polytechnic State University, Bioinformatics Capstone

Undergraduate Bioinformatics Consultant (Internship), April 2025 – June 2025

Advisors: Jean Davidson, Paul Anderson, John Butler

- Identified 36 candidate biomarkers of pre-metastatic tumor buds using triplet RNA-seq data comparing tumor bud, carcinoma, and stromal compartments across multiple patients
- Developed a modular, interactive RNA-seq analysis pipeline in R. Built core modules for data cleaning, differential gene expression analysis, PCA, GSEA, and GO analysis

Jazz Pharmaceuticals

Regulatory Strategy Intern, June 2023 – August 2023

- Conducted a comprehensive review of real-world evidence (RWE) use in FDA NDAs, BLAs, and supplemental submissions, examining purposes of use, primary endpoints, and cases where submissions were unsuccessful due to RWE control arm limitations

RELEVANT PROJECTS

VIDA: Visual Intelligence for Detecting Anemia, California Polytechnic State University, Deep Learning (CSC 487)

- Co-developed an ensemble model approach for anemia detection from peripheral blood smear images and CBC data. Fine-tuned VGG-16 architecture on images, achieving an accuracy of 80.18%, which increased to 85.02% with the addition of CBC data. Assessed model performance using Grad-CAM, identifying prediction drivers and visual cues that influenced model decisions

SKILLS

Programming: Python, R, Java, Bash, HTML, CSS

Machine Learning & Data Science: PyTorch, scikit-learn, pandas, NumPy

Bioinformatics Analysis: Bulk RNA-seq, single-cell RNA-seq, ATAC-seq, whole-genome sequencing

Bioinformatics tools: Seurat, Scanpy, Bioconductor, GATK, bcftools, BLAST, Clustal Omega, AlphaFold

Computing & Visualization: Linux, HPC (LSF), Conda, Git, Shiny, ggplot2, ImageJ

Wet-lab Techniques: PCR/RT-PCR, DNA/RNA extraction and purification, gel electrophoresis, aseptic technique, microbial staining and plating

SERVICE

Alzheimer's Association

Walk to End Alzheimer's Volunteer, 2021-2024

- Assisted with team fundraising events, contributing to over \$60,000 raised annually (2022–2024), and supported annual Walk to End Alzheimer's operations